NUCLEAR POWER SAFETY FAKE NEWS DETECTOR

Comprehensive Technical & System Explanation Report (UK & IAEA Scope)

1. Executive Summary

This system is an AI-powered verification platform designed to detect fake news, prevent public panic, and compute a transparent **0–100 Trust Score** for statements regarding nuclear power safety. Operating under UK regulatory scope and IAEA international standards, it processes **3 input modalities** (raw text, article links/URLs, and news screenshots/images) to deliver defensible decision support.

2. Core Concepts Explained in Simple Terms

- **Natural Language Processing (NLP):** AI branch that allows computers to read, clean, and analyze human language text in simple single-line steps.
- **Machine Learning (ML):** Systems that learn predictive news patterns directly from thousands of training examples instead of relying on manual rules.
- **Computer Vision (OCR):** Technology that scans news images and converts visual printed pixels into digital readable text.

3. Models Used in the Project & Their Roles

Model Name	Type & Framework	Role & Explanation in Simple Terms	Data Consumed
RoBERTa / DeBERTa	Transformer Deep Learning (PyTorch)	Main Text Classifier: Evaluates linguistic context to detect subtle fake news patterns versus authentic technical reporting.	Fine-tuned on 654 UK & IAEA nuclear safety claims.
EasyOCR	Computer Vision OCR (PyTorch + OpenCV)	Image Text Extractor: Reads printed text from uploaded news screenshots and social media graphics.	Raw image pixels (PNG, JPG, WebP screenshots).
VADER & TextBlob	NLP Sentiment & Tone Analyzer	Emotional Tone Checker: Measures whether article phrasing is calm/objective or overly sensational/alarmist.	Parsed article text & headline tokens.
Domain Reputation Engine	Heuristic Rule Evaluator	Publisher Verifier: Assigns credibility rankings based on domain authority (e.g., GOV.UK, BBC News vs clickbait blogs).	URL domain names & publisher headers.
Fact Corroboration Engine	Keyword Overlap & Entity Matcher	Official Log Cross-Checker: Verifies statements against authenticated nuclear safety incident databases.	IAEA IEC, UK ONR & EURDEP reference claims.
Multi-Factor Trust Engine	Ensemble Decision Engine	Score Calculator: Combines model confidence, publisher trust, corroboration, and tone into a final 0–100 Trust Score.	Outputs from all 5 sub-models above.

4. Datasets & Domain Scope

- **Primary Fine-Tuning Dataset (data/sample_dataset.csv):** 654 unique rows focusing on UK nuclear power stations (*Hinkley Point C, Sizewell B, Torness, Heysham, Dungeness B, Sellafield, Wylfa*) and IAEA standards.
- **Held-Out Test Dataset (data/holdout_test.csv):** 30 hand-crafted test statements never seen during training to evaluate real-world generalization (achieving 93% accuracy).
- **Data Specificity:** Pure UK and IAEA scope with zero non-UK or Indian data included.

5. Data Sources Specific to Nuclear Fact-Checking

1. **IAEA Incident and Emergency Centre (IEC) Reports:** Official international bulletins for radiological incidents.
2. **National Regulators:** UK Office for Nuclear Regulation (ONR), UKAEA, DESNZ / GOV.UK, and US NRC declarations.
3. **Radiation Monitoring Networks:** EURDEP (European Radiological Data Exchange Platform) & EPA RadNet.
4. **Nuclear Fact-Checking Archives:** Verified tagged archives from Full Fact UK, Snopes, and PolitiFact.

6. Handle Uncertainty Honestly (Section 7 Policy)

For high-consequence topics like nuclear safety, when a statement has no corroborating match in official logs and lacks clickbait/alarmist markers, the system **does not force a fake score**. It flags the item as **'Insufficient Verification Data'** (Trust Score 50–65/100) to prevent dangerous false positives on genuine safety information.

7. Trust Score Calculation Weights

- **45% Model Confidence:** Transformer classifier prediction probability.
- **25% Source Credibility:** Reputation ranking of the news domain/source.
- **20% Fact Corroboration:** Match against verified IAEA IEC and UK ONR safety logs.
- **10% Tone Neutrality:** Absence of clickbait / sensational emotional language.